

Human gut microbiota: does diet matter?



The human oro-gastrointestinal (GI) tract is a complex system, consisting of oral cavity, pharynx, oesophagus, stomach, small intestine, large intestine, rectum and anus, which all together with the accessory digestive organs constitute the digestive system. The function of the digestive system is to break down dietary constituents into small molecules and then absorb these for subsequent distribution throughout the body. Besides digestion and carbohydrate metabolism, the indigenous microbiota has an important influence on host physiological, nutritional and immunological processes, and commensal bacteria are able to modulate the expression of host genes that regulate diverse and fundamental physiological functions. The main external factors that can affect the composition of the microbial community in generally healthy adults include major dietary changes and antibiotic therapy. Changes in some selected bacterial groups have been observed due to controlled changes to the normal diet e.g. high-protein diet, high-fat diet, prebiotics, probiotics and polyphenols. More specifically, changes in the type and quantity of non-digestible carbohydrates in the human diet influence both the metabolic products formed in the lower regions of the GI tract and the bacterial populations detected in faeces. The interactions between dietary factors, gut microbiota and host metabolism are increasingly demonstrated to be important for maintaining homeostasis and health. Therefore the aim of this review is to summarise the effect of diet, and especially dietary interventions, on the human gut microbiota. Furthermore, the most important confounding factors (methodologies used and intrinsic human factors) in relation to gut microbiota analyses are elucidated.